

Greta Brown

brow6802@umn.edu • gretab5802.github.io

EDUCATION

University of Minnesota, Twin Cities

Minneapolis, MN

College of Science and Engineering

Fall 2025

Bachelor of Science in Computer Science

GPA: 3.8

University of Minnesota, Twin Cities

Minneapolis, MN

College of Science and Engineering

Expected Spring 2027

Masters of Science in Computer Science

SKILLS

Technical Skills: C/C++, Python, Java, OCaml, R, MatLab, HTML, CSS, JavaScript, TypeScript, Git, Arduino, Raspberry Pi, ROS/ROS2, Linux, Public Speaking

Language: Fluent in English; Conversational Proficiency in Spanish

RELEVANT EXPERIENCE

NASA Ames Research Center

Moffett Field, California

Planetary Sampling Automation Development Intern

August 2025 - Present

- Conduct research on autonomous extraterrestrial planetary sampling using a robotic drilling system.
- Analyze drilling telemetry data using machine learning and change-point detection methods.
- Prepared simulant regolith for experiments, operated drilling system, and collected audio and sensor data.
- Integrated fault detection systems with drilling platform using ROS 2 middleware.

University of Minnesota - Twin Cities Next Generation Robotics Laboratory

Minneapolis, MN

Research Assistant

August 2024 - Present

- Conduct research in field robotics and autonomous systems, focusing on real-world system behavior, sensing, and experimental validation.
- Participate in weekly research discussions, presenting literature reviews and updates on ongoing experimental work and system development.

University of Minnesota - Twin Cities Department of College of Science and Engineering

Minneapolis, MN

Undergraduate Teaching Assistant

January 2023 - January 2026

- Led weekly lab sessions and office hours for Java and Python courses, providing coding support and reinforced programming and AI concepts.
- Graded assignments and exams with detailed feedback.

University of Texas at Austin - Oden Institute for Computational Engineering and Sciences

Austin, TX

Robotics and Autonomous Systems Research Intern

June 2025 - August 2025

- Designed and validated reinforcement learning policies for quadruped locomotion on the Unitree Go2 in simulation environments.
- Developed interfaces for low-level control and sensor data handling for quadruped systems, focusing on sim-to-real transfer and real-world deployment.

AWARDS

CRA Outstanding Undergraduate Researcher Award Honorable Mention

2025-2026

- Research in autonomous planetary robotics, focusing on field-deployable drill perception and planning.

University of Minnesota - Twin Cities

Undergraduate Research Opportunities Program (UROP) Award

August 2024 - December 2024

- Researched multi-agent real robot systems using reinforcement learning and embodied evolution.
- Developed a novel hardware-software integration method for the most current platforms available.